

ER Diagram

About

The **House Rent Management System** is a web-based application that enables users to search, list, and manage rental properties efficiently. It supports three types of users: **Renter (Tenant)**, **Owner (Landlord)**, and **Admin**. The system allows renters to book property visits, owners to manage property listings, and administrators to monitor and control the platform.

Key Features

- User Registration and Secure Login
- Role-Based Access (Renter, Owner, Admin)
- Add, Update, and Delete Property Listings
- Search and Filter Properties
- Property Booking Management
- Booking Status Tracking
- Review and Rating System
- Wishlist Management
- Admin Approval for Property Owners
- Responsive MERN Stack Application

User Roles

1. Renter (Tenant)

The **Renter** searches for rental properties and submits booking requests.

Responsibilities

- Register and log in.
- Browse available properties.
- View property details.
- Book a property visit.
- Track booking status.
- Write reviews after visiting.
- Save favorite properties to the wishlist.

2. Owner (Landlord)

The **Owner** lists and manages rental properties after receiving approval from the Admin.

Responsibilities

- Register as an owner.
- Receive Admin approval.
- Add new property listings.
- Edit property information.
- Delete property listings.
- Update property availability.
- View booking requests.
- Approve or reject bookings.

3. Admin

The **Admin** manages the complete application.

Responsibilities

- Verify and approve owner accounts.
- Manage all users.
- Monitor property listings.
- Approve or reject reported listings.
- Maintain platform security.
- Ensure smooth system operation.

Entity Description

1. User

Purpose: Stores information about all users of the system.

Attributes

- User_ID (Primary Key)
- Name
- Email
- Password
- Phone

- Role (Renter / Owner / Admin)
- Profile_Image
- Created_At

2. Property

Purpose: Stores all rental property details added by owners.

Attributes

- Property_ID (Primary Key)
- Title
- Description
- Address
- City
- State
- Price
- Property_Type
- Bedrooms
- Bathrooms
- Area
- Amenities
- Images
- Status
- Availability
- Furnished
- Parking
- Pets_Allowed
- Virtual_Tour_URL
- Views
- Owner_ID (Foreign Key)

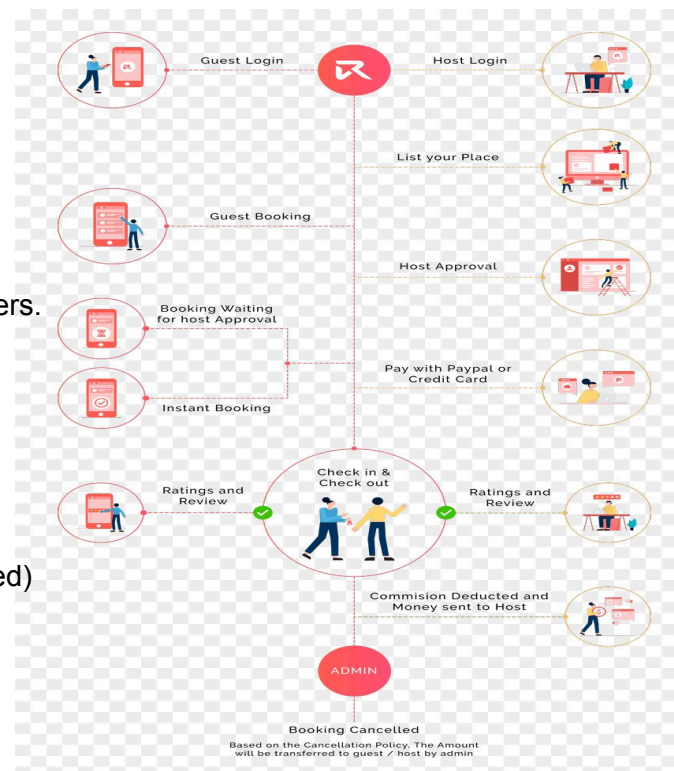


3. Booking

Purpose: Stores property booking requests made by renters.

Attributes

- Booking_ID (Primary Key)
- Visit_Date
- Visit_Time
- Status (Pending / Approved / Cancelled / Completed)
- Message



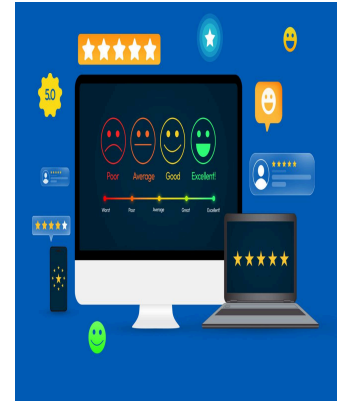
- Tenant_ID (Foreign Key)
- Property_ID (Foreign Key)
- Created_At

4. Review

Purpose: Stores ratings and feedback provided by renters for properties.

Attributes

- Review_ID (Primary Key)
- Rating (1–5)
- Comment
- User_ID (Foreign Key)
- Property_ID (Foreign Key)
- Created_At

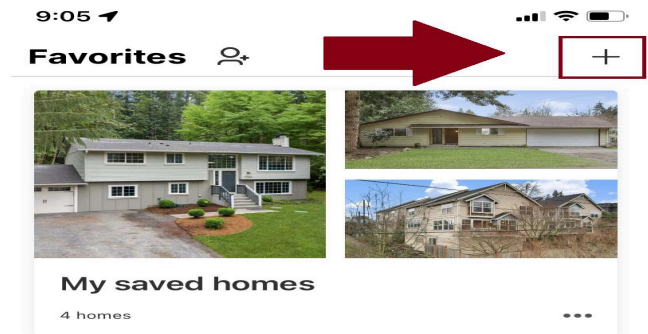


5. Wishlist

Purpose: Stores properties saved by users for future reference.

Attributes

- Wishlist_ID (Primary Key)
- User_ID (Foreign Key)
- Property_ID (Foreign Key)
- Added_At



Relationship Summary

- One **User** can own **many Properties**.
- One **User** can make **many Bookings**.
- One **Property** can have **many Bookings**.
- One **User** can write **many Reviews**.
- One **Property** can receive **many Reviews**.
- One **User** can save **many Properties** in the **Wishlist**.
- One **Property** can appear in the wishlists of multiple users.

Relationship Explanation

Relationship	Cardinality
User → Property	One-to-Many (1:N)
User → Booking	One-to-Many (1:N)
Property → Booking	One-to-Many (1:N)
User → Review	One-to-Many (1:N)
Property → Review	One-to-Many (1:N)
User → Wishlist	One-to-Many (1:N)
Property → Wishlist	One-to-Many (1:N)

